$$X_{11} + X_{12} + X_{13} = 1 \quad \text{(build only one treatment plant)} \quad ...[7.10]$$

$$X_{21} + X_{22} + X_{23} = 1 \quad \text{(build only one treatment plant)} \quad ...[7.11]$$

$$X_{31} + X_{32} + X_{33} = 1 \quad \text{(build only one treatment plant)} \quad ...[7.12]$$

...

7.2 IRRIGATION EXAMPLE

An irrigation project is being proposed (refer to Figure 7.2). Two sources of water are possible: surface water from a reservoir and pumped groundwater. Two possible sizes of reservoir are to be considered, and for environmental reasons a minimum stream flow is required below the reservoir. However, return flow from irrigation can help meet that requirement. Data on water demand, return flows, capacities, and costs are provided in Tables 7.2 and 7.3. Assume that crop revenue minus non-water costs is $280/acre/year.

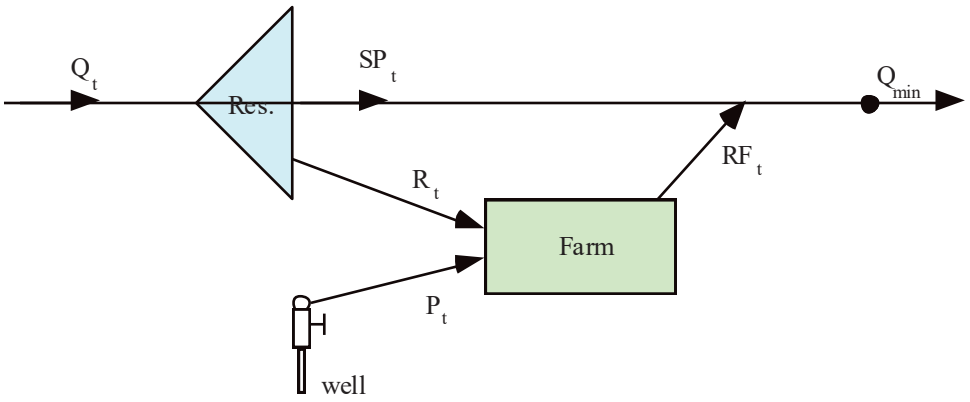


Figure 7.2: Irrigation Supply from Reservoir and Well Water

Table 7.2: Water Demand and Hydrologic Data

Season	Q_t (acft)	Water Demand (acft/acre)	Q_{min} (acft/day)	Return Flow, percent
1--8 mo.	200	3.0	0.5	50
2--4 mo.	600	1.5	0.5	30

Table 7.3: Facility Capacities and Costs

Facility	Costs		Capacity
	Capital, $/year	Operation and Maintenance, $/acft	
Reservoir A	5000	3	350 acft
Reservoir B	15000	3	700 acft
Pump	1500	15	2 acft/day

Develop an MIP model to determine the optimal size of farm, size of reservoir, and whether to build a well.

Solution:

The model structure will be:

Max Z = Benefits - Costs ...[7.13]

s.t.: (1) Mass Balance in Reservoir ...[7.14]

(2) Water Demand ...[7.15]

(3) Reservoir Storage ≤ Capacity ...[7.16]

(4) Pump Flow ≤ Capacity ...[7.17]

(5) Minimum stream flow ≥ 0.5 acft/day ...[7.18]

Define the following variables:

- Size of farm in acres = AC
- Reservoir integer variables I_A, I_B = 0 or 1
- Pump integer variable = I_p
- Assume constant but unknown carryover storage in the reservoir (i.e., S1 = S3)
- Releases from the reservoir, R_t, and pumping from the well, P_t, are in acft/season

The detailed model can be written as:

$$\text{Max } Z = 280\ AC - 5000\ I_A - 15000\ I_B - 1500\ I_p - 3\ (R_1 + R_2) - 15\ (P_1 + P_2) \quad ...[7.19]$$

$$\text{s.t.:} \quad S_2 = S_1 + 200 - SP_1 - R_1 \quad ...[7.20]$$

$$S_1 = S_2 + 600 - SP_2 - R_2 \quad ...[7.21]$$

$$R_1 + P_1 \geq 3\ AC \quad ...[7.22]$$

$$R_2 + P_2 \geq 1.5\ AC \quad ...[7.23]$$

$$S_1 \leq 350\ I_A + 700\ I_B \quad ...[7.24]$$

$$S_2 \leq 350\ I_A + 700\ I_B \quad ...[7.25]$$

$$P_1 \leq 2\ (8)\ 30\ I_p \quad ...[7.26]$$

$$P_2 \leq 2\ (4)\ 30\ I_p \quad ...[7.27]$$

$$SP_1 + 0.5\ (R_1 + P_1) \geq 0.5\ (8)\ 30 \quad ...[7.28]$$

$$SP_2 + 0.3\ (R_2 + P_2) > 0.5\ (4)\ 30 \quad ...[7.29]$$

Note that if I_A and $I_B = 0$, then S_1 and $S_2 = 0$, so mass balance is correct with either one or no reservoirs.

7.3 INVESTMENT TIMING PROBLEM

Another type of problem where integer programming is very efficient is in investment timing. Consider a water supply utility in which supply capacity can be added by any of five independent projects. The additional supply and the project costs (already been discounted to present value) are shown in Table 7.4. Possible discrete construction periods and related growth in demand are also summarized in the table. Assume that end-of-period added capacities shown must be constructed at the beginning of each 5-year period because projects can be constructed only during year 1, 6, 11, or 16.

Table 7.4: Project Scheduling Problem Data

Period	Years	Discounted Cost of Additional Capacity Units of Additional Capacity					Additional Capacity Required at the End of Period
		2	4	6	8	10	
1	1-5	12	15	18	23	26	2
2	6-10	8	11	13	15		6
3	11-15	6	8				8
4	16-20	4					10

[a]This is the total required capacity that must be added to the existing capacity at the beginning of year 1.